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Research article



Observed nitrogen accumulation during salinity and temperature surges in a coastal dam controlled lake

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ABSTRACT

In this study, we tested the hypothesis that bidirectional dam operations modulate Total Nitrogen (TN) accumulation through changes in salinity, temperature, and hydrodynamic flushing in Guana Lake, Florida. We conducted one year of monthly surveys, complemented by continuous and discrete measurements of TN and key water quality parameters (salinity, temperature, pH, chlorophyll-a, and dissolved oxygen).

During summer 2024, management actions increased saltwater inflow to suppress freshwater vegetation, which reduced net outflow and tidal flushing. Together with high evaporation, these actions coincided with elevated salinity and temperature, exceeding values in the adjacent estuary. These conditions were associated with increased organic nitrogen and ammonium concentrations, elevated chlorophyll-a, and reduced dissolved oxygen, consistent with altered nitrogen cycling pathways. Nitrate and nitrite remained below detection limits throughout the study period.

Our results indicate that episodic salinity and temperature increases were associated with TN accumulation, particularly in reduced and organic forms, under the observed management regime. However, this study did not directly quantify process rates or sediment fluxes, and mechanistic interpretations are based on observed correlations and established biogeochemical frameworks. The findings suggest that dam operations that modify residence time and flushing may influence nitrogen accumulation patterns, but quantitative evaluation of alternative management scenarios requires further hydrodynamic and biogeochemical modeling.

This study provides observational evidence of coupled management, salinity, and nitrogen dynamics in a dam-managed coastal lake. While system-specific hydrologic and geomorphic conditions must be considered, the conceptual framework may inform future investigations of nitrogen dynamics in similar managed estuarine systems.

1. Introduction

Rising temperatures increase evapotranspiration in closed water bodies such as reservoirs and artificial lakes (J. Wang et al., 2022). In brackish environments, enhanced evapotranspiration elevates salinity (Sumner and Belaineh, 2005). Increased salinity exposure in tidal wetlands reduces dissolved oxygen (DO) availability and elevates sulfide concentrations in sediments, thereby altering nitrogen cycling processes (Osborne et al., 2015). Higher salinity can enhance dissimilatory nitrate reduction to ammonium (DNRA) and nitrogen fixation, which recycle or introduce new nitrogen into aquatic ecosystems (Xie et al., 2020; Huang et al., 2024). In addition, higher temperatures reduce oxygen solubility, further affecting aquatic biogeochemical processes (Tromans, 1998; Xing et al., 2014). Elevated salinity and temperature can suppress nitrification and denitrification, potentially reducing sediment nitrogen retention and increasing nitrogen export to the water column (Wang et al., 2022; Torregrosa-Crespo et al., 2023).

Previous studies have quantified the influence of salinity and

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temperature on nitrogen transformation processes. Salinity and temperature strongly affect nitrogen mineralization, with higher salinity and temperature promoting ammonium accumulation (Gao et al., 2014). Increased salinity can also enhance ammonium and ammonia uptake by phytoplankton and promote the release of dissolved organic nitrogen (DON) during senescence (Kumar et al., 2018; Bronk et al., 1994). In systems with variable flushing, nitrogen retention generally decreases as flushing rates increase. Damming rivers and lakes alters hydrodynamics and sediment transport, reshaping nitrogen cycling within aquatic systems (Liermann et al., 2012). Reduced flow velocity near dams promotes sedimentation of fine particles with high adsorption capacity, increasing the retention of total nitrogen (TN) and organic carbon (Winton et al., 2019). Conversely, rapid flushing can export nitrogen and reduce eutrophication risk, whereas poorly flushed systems are more influenced by recycled nitrogen and more susceptible to eutrophication (Monbet, 1992; Cloern, 2001; Philips et al., 2004; Walve et al., 2021).

Tidal water-level fluctuations further influence nitrogen dynamics through cyclic drying and rewetting of sediments, which alters oxygen availability and substrate supply and accelerates coupled nitrification–denitrification processes (Thouvenot-Korppoo et al., 2009). Sediment nitrogen fluxes can represent a substantial component of the nutrient budget, and sediment oxygen demand is often correlated with dissolved inorganic nitrogen release (Fisher et al., 2005; Larson et al., 2020). External nitrogen inputs, including atmospheric deposition, watershed inflow, and biological nitrogen fixation, have increased substantially over the past century and can strongly influence reservoir nitrogen budgets (Zhu et al., 2018; Horváth et al., 2013). Total nitrogen in aquatic systems includes both inorganic and organic forms, which differ in bioavailability and ecological impact (Zhao et al., 2021).

Water-level and flow management in reservoirs has been shown to influence nutrient dynamics and water quality. For example, controlled releases in Lake Okeechobee reduced phosphorus inputs (Tarabih et al., 2023), adaptive management in the Three Gorges Reservoir improved ecological outcomes (Hu et al., 2014), and modified operations in Saemangeum Lake affected dissolved oxygen dynamics (Kwak et al., 2023). However, most existing studies focus on freshwater systems, whereas nitrogen cycling in brackish and saline environments differs fundamentally due to salinity-driven biogeochemical and microbial controls (Wei et al., 2023). In dam-managed brackish lakes, the combined effects of bidirectional saltwater exchange, residence time variability, and episodic salinity intrusion on nitrogen accumulation remain poorly quantified.

The influence of dam operations on nitrogen cycling in brackish coastal lakes, particularly under episodic saltwater intrusion and variable flushing conditions, is not well understood. In particular, it remains unclear how management-driven salinity and hydrodynamic variability interact to control TN accumulation and speciation.

The objective of this study is to quantify how bidirectional dam operations influence nitrogen accumulation and speciation in a dam-managed brackish coastal lake. We tested the hypotheses that: (i) episodic increases in salinity and temperature are associated with increased TN accumulation, particularly in reduced and organic forms; (ii) reduced tidal flushing enhances nitrogen retention within the lake; and (iii) spatial variability in salinity dynamics controls spatial patterns of TN concentrations.

Guana Lake, Florida, is a dam-managed brackish system influenced by tidal exchange, freshwater inflow, and urban runoff. We present results from a one-year field campaign (October 2023–October 2024) examining how dam operations, salinity, and temperature variability relate to TN concentrations, dissolved oxygen, chlorophyll-a, and pH.

2. Methods

2.1. Study area

Guana Lake (hereafter referred to as the "Lake") is a 9.5 km^2 impounded riverbed within the Guana Tolomato Matanzas (GTM) Estuary in Northeast Florida (Fig. 1a), receiving freshwater through a weir from the higly urbanized Ponte Vedra area in the north and saltwater via tidal exchange from the Guana River (hereafter referred to as the "River") to the south. A water control structure, the Guana Dam, separates the natural River from the impounded Lake and regulates the water flux between the Lake and the estuary (Fig. 2b). In general, the Lake displays a dynamic salinity gradient ranging from 0 to 40 PSU, forming distinct ecological zones. From Mickler's Weir to station GL2, water is shallow and fresh. Between GL2 and Lake Middle, conditions are brackish, while from Lake Middle to the Guana Dam, salinity levels are similar to those of the River. However, these conditions vary seasonally and are influenced by regulating saltwater inflow at the Guana Dam.

A six-year collaborative effort involving the GTM National Estuarine Research Reserve (GTMNERR) has shown that the GTM Estuary has elevated nutrients and chlorophyll-a throughout the entire estuary, but particularly in the upper-middle portions, and that internal processes like sediment recycling and vegetation growth/senescence are important drivers of nutrient availability (Dix et al., 2025). Additionally, US EPA (2012) indicates that nitrogen concentrations, often peaking in the northern region of Guana Lake, frequently exceed the state threshold for Class III estuarine waters (0.65 mg/L, 62-302.700 F.A.C.), indicating the combined influence of reservoir operations and nutrient enrichment. Reservoir operations can influence nutrient release by affecting hydrodynamic and biogeochemical processes (Xu et al., 2021). Urban runoff from the northern watershed delivers high nutrient loads and algal biomass to the Lake (Grogan et al., 2023). Additionally, the Guana River watershed includes managed landscapes such as golf courses upstream

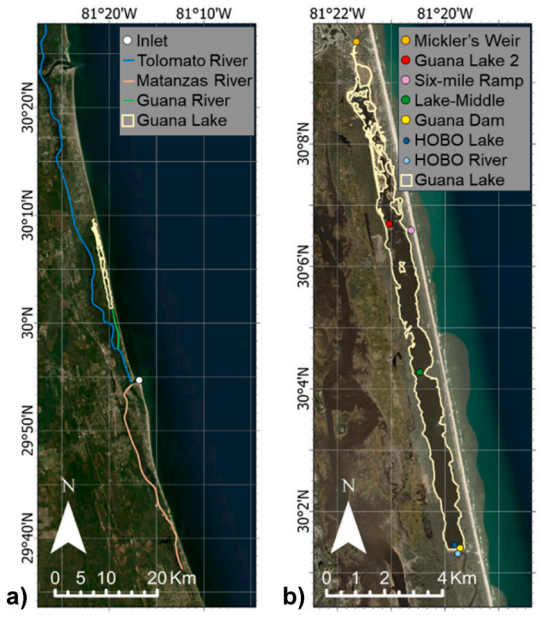


Fig. 1. (a) Guana–Tolomato–Matanzas (GTM) estuary, with the Tolomato River marked in blue, the Matanzas River in orange, and the Guana River, draining Guana Lake, highlighted in green. The white outline traces the Lake's contour. (b) Guana Lake at higher spatial resolution (outlined in white). Dots indicate the locations of the monitoring stations: Mickler's Weir (orange), the GL2 station in Guana Lake (red), the Lake Middle (LM) station (green), and the Guana Dam (coincident with the sampling station Lake South, or LS, in yellow). The Six-Mile Ramp location (pink dot) and the location of the two HOBO sensors discussed in Section 2.2 are also shown.

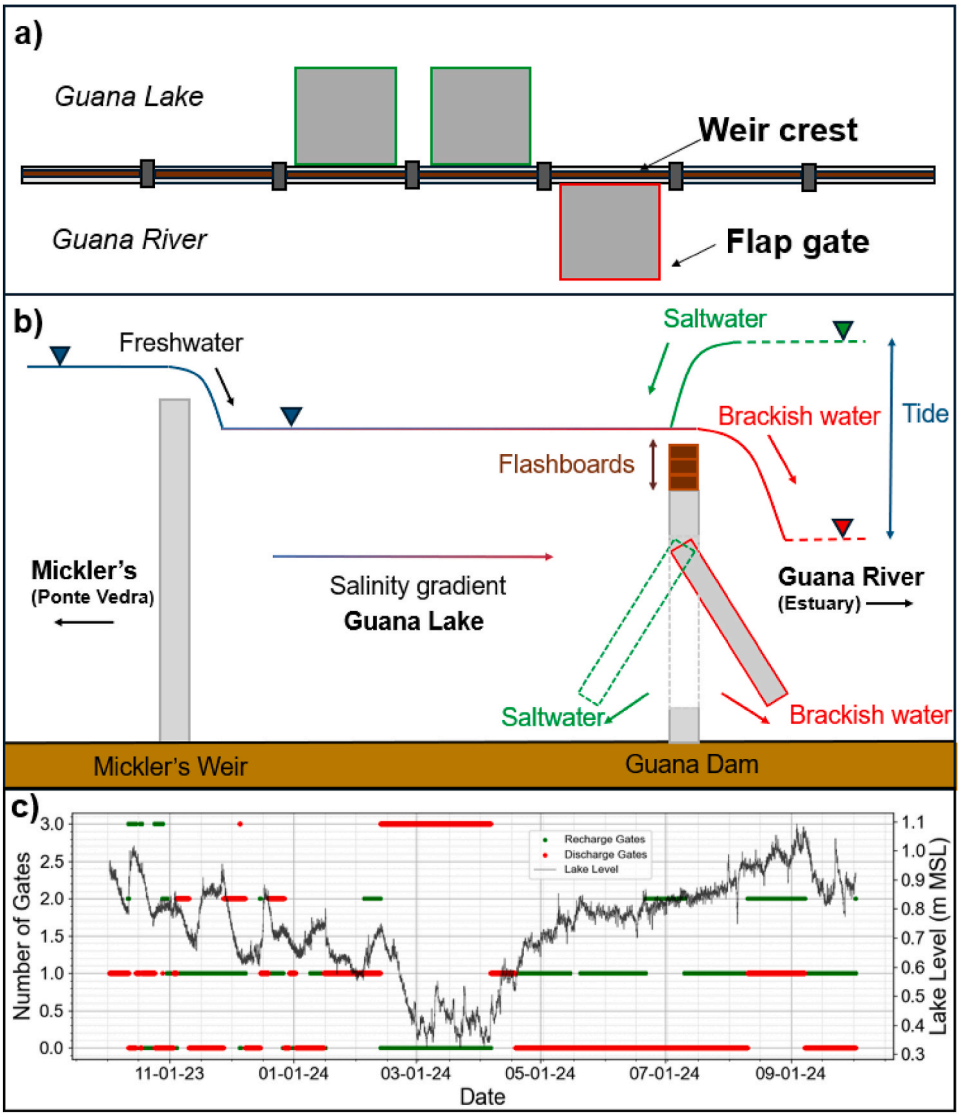


Fig. 2. Schematic Guana Dam operating configurations. (a) Plan view with two gates admitting flow to the lake (outlined in green) and one gate discharging to the River (outlined in red). (b) Side view of Guana Lake and the two water control structures. In general, water flows from the left side (Mickler's, Ponte Vedra) to the right side of the figure (Guana Dam). (c) Time series of gate configurations (red and green) and water level in the Lake.

of the lake, which represent potential sources of nutrient inputs through fertilizer application. Excessive nutrient loading in aquatic ecosystems may result in eutrophication and increase the proliferation potential of phytoplankton and macroalgae (Valiela et al., 1997; Heisler et al., 2008; Yu et al., 2018).

2.2. Data collection and analysis

During monthly field surveys from October 2023 to October 2024, we collected water quality data at Guana Lake using a combination of in situ instrumentation, continuous and discrete sampling. In situ measurements included temperature, DO, chlorophyll-a, pH and salinity, collected with a YSI ProDSS (Xylem). Additionally, four stations were selected for discrete sampling: "Mickler's Weir" (MK), "Guana Lake 2" (GL2), "Lake Middle" (LM), and "Lake South" (LS, coincident with the location of the Guana Dam), as shown in Fig. 1b.

At each station, 500 mL grab samples were collected in acid-washed, dark, sterilized HDPE bottles. Samples were taken at 0.2 m depth. The lake is shallow (ranging 0.4 m – 1.2 m) and generally well mixed, with is no evidence of strong vertical stratification. Sampling followed the Florida Department of Environmental Protection Standard Operating Procedures (FDEP, 2017). One unfiltered and one 0.45 μm filtered and acidified (< pH 2 with H_2SO_4) subsample were preserved on ice and transported to the laboratory at 4 °C for analysis of total nitrogen, including the sum of nitrate and nitrite ($NO_3^- + NO_2^-$, hereafter NOx), ammonium (NH_4^+), and Total Kjeldahl Nitrogen (TKN). Analyses were performed by the IFAS laboratory (University of Florida) using the methods (U.S. EPA, 2012): NOx – EPA Method 353.2 NH_4^+ – EPA Method 350.1 TKN – EPA Method 351.2.

DON was determined by subtracting NH_4^+ from filtered TKN (TKNF), following the standard nomenclature where DON refers to organic nitrogen passing through a 0.45 μm-filter. DIN was calculated as the sum of NOx and NH_4^+. TN was computed by adding TKN (which includes organic nitrogen and NH_4^+) and NOx, hence reflecting the formula TN = TKN + NOx. Particulate nitrogen (PN) was derived by subtracting the sum of DON and DIN from TN, isolating the nitrogen fraction bound to particles. In many cases, NOx concentrations were below detection limits and therefore contributed minimally to DIN and TN calculations. Method detection limits and QA/QC procedures were provided by the IFAS laboratory following U.S. EPA protocols; concentrations below detection limits (0.05 mg/L) were treated as half the detection limit for statistical analyses.

Data for continuous monitoring of temperature, DO and salinity were also collected with a YSI EXO2 (YSI) deployed at Mickler's (orange dot in Fig. 2b) placed by the GTMNERR staff (data and metadata available at https://floridaapdata.org/). All the sensors were calibrated and deployed according to the manufacturer recommendations and Florida Department of Environmental Protection (FDEP). Finally, two HOBO Onset U24 (HOBO) conductivity-temperature sensors were deployed near the Guana Dam, one inside the Lake, and one in the River, respectively. The one deployed in the Lake was paired with a HOBO water level sensor. Continuous measurements were recorded at 15 min intervals. For comparison with discrete samples, continuous data were temporally aggregated to daily means and matched to discrete sampling dates using aligned timestamps.

Pairwise associations among TN, salinity and temperature indicators were quantified using zero-lag Pearson correlation coefficients (r), computed on dates when both variables were available. Pearson's r quantifies the strength and direction of linear covariation, ranging from -1 (perfect negative association) to $+1$ (perfect positive association), with values near zero indicating weak or no linear relationship.Finally, the associated p-value was obtained from a two-tailed significance test. Statistical significance was evaluated using a threshold of $p < 0.05$. For each station, r was reported together with the number of paired observations (n). All statistical analyses were performed in Python.

2.3. Guana Dam management

The water level of Guana Lake is regulated by the bidirectional Guana Dam. The primary objective of water level management in the Lake is to sustain high-quality wetland habitats for a wide range of native wildlife that depend on brackish and freshwater marshes throughout their life cycles. Unlike naturally fluctuating wetlands shaped by seasonal rainfall and drought, impoundments like Guana Lake require strategic manipulation to mimic these dynamic conditions. Without such variation, stagnant water and constant salinity levels can promote undesirable vegetation, muck accumulation, and overall habitat degradation. The dam is equipped with seven aluminum gates that can either release brackish water to the estuary or allow saltwater intrusion during high tides from the Guana River (discharge and recharge mode; highlighted in red and green in Fig. 2, respectively). Additionally, the weir crest height can be adjusted with 12.7 cm wooden flashboards to regulate the Lake level gradually and precisely. By varying the number of active gates and flashboards, the Lake level is changed, with lower levels providing flood control and higher levels providing recreational benefits. As the terminal drainage point of the Ponte Vedra watershed, the dam helps convey excess rainfall during hurricane season and prevents storm surge propagation upstream, thereby reducing flood risk in developed areas. In general, the dam configuration controls water retention within the basin. Periods of increased discharge toward the estuary could promote flushing, increasing constituent export and reducing residence time, whereas gate closure or recharge mode might enhances retention, longer residence times, and the potential for nitrogen accumulation within the lake. Occasionally, all discharge gates are closed, and one or more recharge gates are left open to drastically increase salinity levels and suppress nuisance freshwater vegetation. This operation was performed in summer 2024 (Fig. 2c), and we describe below how it coincided with changes in nitrogen cycling. Additionally, gate closure and recharge-mode operation are expected to increase residence time by reducing net outflow and bidirectional exchange, consistent with the observed salinity accumulation.

3. Results

3.1. Salinity and temperature

Salinity in Guana Lake gradually increased from north to south during the year (Fig. 3a). In the northern part of the Lake near MK, salinity remained close to 0 PSU throughout most of the year because of freshwater discharge. An increase in salinity occurred between mid-May 2024 and mid-July 2024. This occurred for two reasons: first, a reduction in freshwater input due to the temporary closure of Mickler's Weir by the St. John's County management. This closure was intended to increase water levels in the upstream canal and pond areas interspersed throughout the community of Ponte Vedra, which were experiencing enhanced evaporation. Second, there was an increase in saltwater inflow from the Guana Dam, as all discharge gates were closed, and one open gate was set in “recharge mode” (Fig. 2b) by FWC. This operation was intended to increase saltwater intrusion into the Lake to eliminate nuisance freshwater plant species. As shown in Fig. 2c, lake stage rose concurrently during this period, indicating recharge-only operation in which the lake was receiving saltwater without exporting flow. These management actions were associated with salinity increases up to 28 PSU at the GL2 station, which is located in the northern part of the Lake, past the six-mile ramp (Fig. 1b). This area typically marks the limit of saltwater intrusion, where salinity generally remains below 8 PSU throughout the year (Dix et al., 2025). However, the highest salinity levels in the Lake, around 39 PSU, were recorded at LS, where saltwater from the River enters the Lake. In this part of the Lake, salinity reaches its lowest during the fall season, dropping to 12–15 PSU mainly due to intense tropical rainfall events common in Florida. Notably, the fastest drops in salinity occurred during Hurricane Milton (mid-September 2024) and Hurricane Helene (mid-October 2024).

Water temperature exhibited clear seasonal variation in both the Lake and River (Fig. 3b). Starting in early November 2023, temperatures steadily decreased from around 27 °C, reaching a minimum of about 11 °C in early January 2024. Following this minimum, temperatures gradually rose, peaking around July and August 2024 with values close to 30 °C. Subsequently, temperatures declined again towards October 2024. No significant spatial gradient of temperature was observed. The minimal spatial variation in temperature was due to differences in sampling times during the day. Continuous data between mid-January and March 2024 were unavailable due to sensor memory limitations; however, monthly discrete sampling continued during this period, and these data were used to constrain winter conditions in the analysis. High-frequency variability during this period cannot be resolved, but this gap does not affect the analysis of spring–summer salinity intrusion events.

Overall, temperature trends in the Lake and River closely followed each other, although some differences were noticeable (Fig. 4b). The River exhibited greater fluctuations compared to the more stable Lake temperatures, although this is likely because the sensor in the River was located in a shallow area, making it more sensitive to changes in external air temperature. During the warmer summer months, both locations reached temperatures above 30 °C, but the Lake experienced higher temperatures due to reduced tidal flushing. Indeed, the Lake behaved similarly to a closed water body, leading to temperatures up to 3 °C higher than in the adjacent estuary. This pattern is consistent with restricted exchange through the Guana Dam gates. The minimal discharge occurring as overflow from the spillway is negligible compared to the contribution of the gates (Sanders and Feaster, 2004). Consequently, relatively cooler water from the estuary entered the Lake only when the water level in the estuary exceeded that of the Lake (relatively brief period during each tidal cycle, as the mean water level in the Lake is higher than the mean sea level). These limited inflows, combined with high external temperatures and restricted circulation, likely contributed to higher water temperatures within the Lake. Local temperature variability may also be influenced by shallow depth, fetch, and wind-driven mixing. Shallow areas warm more rapidly, and spatial differences in fetch and shading can modulate heat exchange. However, the consistent Lake–River temperature offset during summer, together with restricted tidal exchange, indicates that reduced flushing was a primary control on elevated lake temperatures. The simultaneous rise in

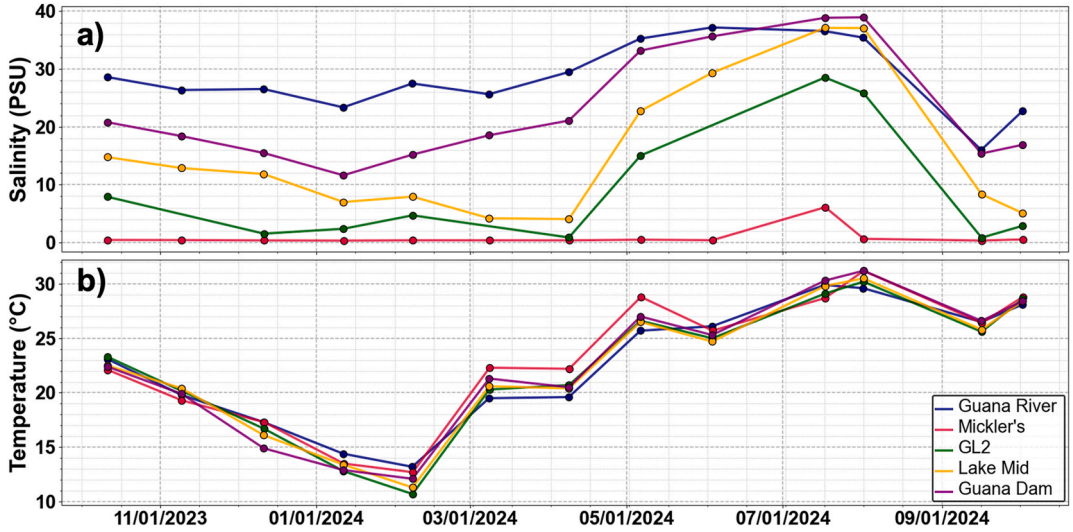


Fig. 3. Time series of (a) salinity and (b) temperature measured at the four stations in Guana Lake and in the adjacent estuary, in the Guana River (location HOBO River, Fig. 1b) by discrete sampling with the YSI proDSS. In both plots, the stations are (from north to south): MK (blue line), GL2 (red line), LM (green line), and LS (yellow line), GR (purple line).

lake level depicted in Fig. 2c further supports recharge-only saltwater inflow under these operating conditions.

Finally, continuous monitoring at the MK station showed how the salinity surge developed in the area (Fig. 4d). During the month of July, two peaks of approximately 12 PSU correspond to the lowest values of DO (Fig. 4e) of the year, coinciding with hypoxic conditions. The coincident peaks in salinity and declines in DO suggest enhanced oxygen demand during the salinity intrusion period, potentially linked to organic matter decomposition, microbial respiration, and altered nitrogen cycling pathways.

3.2. Total nitrogen

Over the year, TN concentrations were higher at the GL2 station (Fig. 5d), which is the area where sediments are intermittently exposed to high-salinity water pulses. The highest TN concentration was observed at the MK station in July 2024 (Fig. 6d), increasing from an annual mean of 1.1 mg L^{-1} to 4.8 mg L^{-1}. While most of the nitrogen occurred as DON, there was a marked surge in ammonium (NH_4^+) (Fig. 6b). This increase was associated with dam management practices that introduce saltwater into the system to suppress excess freshwater vegetation.

The NOx levels are almost zero (Figs. 5c and 6c), which suggests that anthropogenically-derived nitrate from Ponte Vedra (Fig. 1a) may be processed in upstream environments before being transported into the Lake by algal or plant uptake. Lastly, the fraction of particulate nitrogen (PN), calculated as the difference between Total Kjeldahl Nitrogen (TKN) and its soluble fraction (TKNF), also peaks during July 2024 (Fig. 6f). This is consistent with the interpretation that elevated temperatures and salinity contributed to increased PN.

Correlation patterns varied among stations (Table 1; Fig. S1 in the Supplementary Material). The strongest TN–salinity relationship occurred at MK ($r = 0.82$, $p < 0.001$), where salinity increased episodically in response to dam operation and marine intrusion. Because MK normally experiences freshwater conditions, these intrusions produced the largest relative salinity change in the system and coincided with the largest TN increases. At GL2, where conditions remained persistently brackish, the TN–salinity correlation was weaker and not statistically significant ($r = 0.37$, $p = 0.11$). At LS, salinity remained high throughout the monitoring period and TN showed no relationship with salinity ($r = 0.009$, $p = 0.97$). At LM, the non-significant correlation likely reflects mixed water masses and intermediate salinity variability, where competing processes such as dilution, biological uptake, and internal recycling reduce the apparent TN–salinity coupling. These differences indicate that TN responses were most pronounced in the portion of the lake experiencing the largest salinity perturbations, whereas stations with persistently saline or weakly varying conditions showed little or no correlation. Scatter plots illustrating these relationships are provided in Fig. S1 to allow visual evaluation of the correlations.

3.3. Chlorophyll-a, dissolved oxygen and pH

Chlorophyll-a fluorescence signals, expressed in relative fluorescence units (RFU) as a proxy for algal biomass, remained relatively consistent at GL2 and LS, with values generally below 10 RFU throughout most of the year. However, at MK, there was a distinct spike in mid-summer, reaching approximately 60 RFU (Fig. 7d), which indicates a potential localized algal bloom. This increase corresponded to elevated temperatures above 30 °C and higher salinities at that site (12 PSU during the same period, Fig. 4d), suggesting that warm, salt-influenced conditions were consistent with the growth of halotolerant algae. Additionally, continuous monitoring at MK (Fig. 4e) showed DO as low as 0.5 mg/L during the same period. At LM, chlorophyll values also rose notably during the spring, reaching up to 20 RFU. This increase occurred during a period from mid-February to mid-April 2024, when the water level was substantially lowered to allow prescribed burns in the Lake. This burning process could have released nutrients and increased light availability, both of which could have stimulated periphytic and phytoplankton biomass in the short term (Joel et al., 2016). Chlorophyll-a did not show a clear spatial gradient across the Lake (Fig. 7c), except in the Guana Dam area, where elevated tidal flushing consistently reduces concentrations of organic and inorganic matter.

Over the course of the year, pH and DO exhibited pronounced spatial and seasonal patterns across the four stations. Mickler's consistently recorded the lowest pH and DO, with a median pH near 7.5 and frequent dips below 7.6 (Fig. 7a), while the other locations maintained higher, more stable pH values generally between 7.8 and 8.4. DO at MK often fell below 5 mg/L and occasionally near hypoxic thresholds, whereas the other sites typically exceeded 7 mg/L. These patterns suggest greater biogeochemical variability at MK; however, given the monthly sampling resolution, potential drivers such as localized blooms or mixing effects cannot be resolved and are therefore proposed only as possible explanations. For TN–DO, correlations are weak to moderately negative ($r \approx -0.18$ to -0.33) at GL2, LS, and MK, indicating that higher TN is

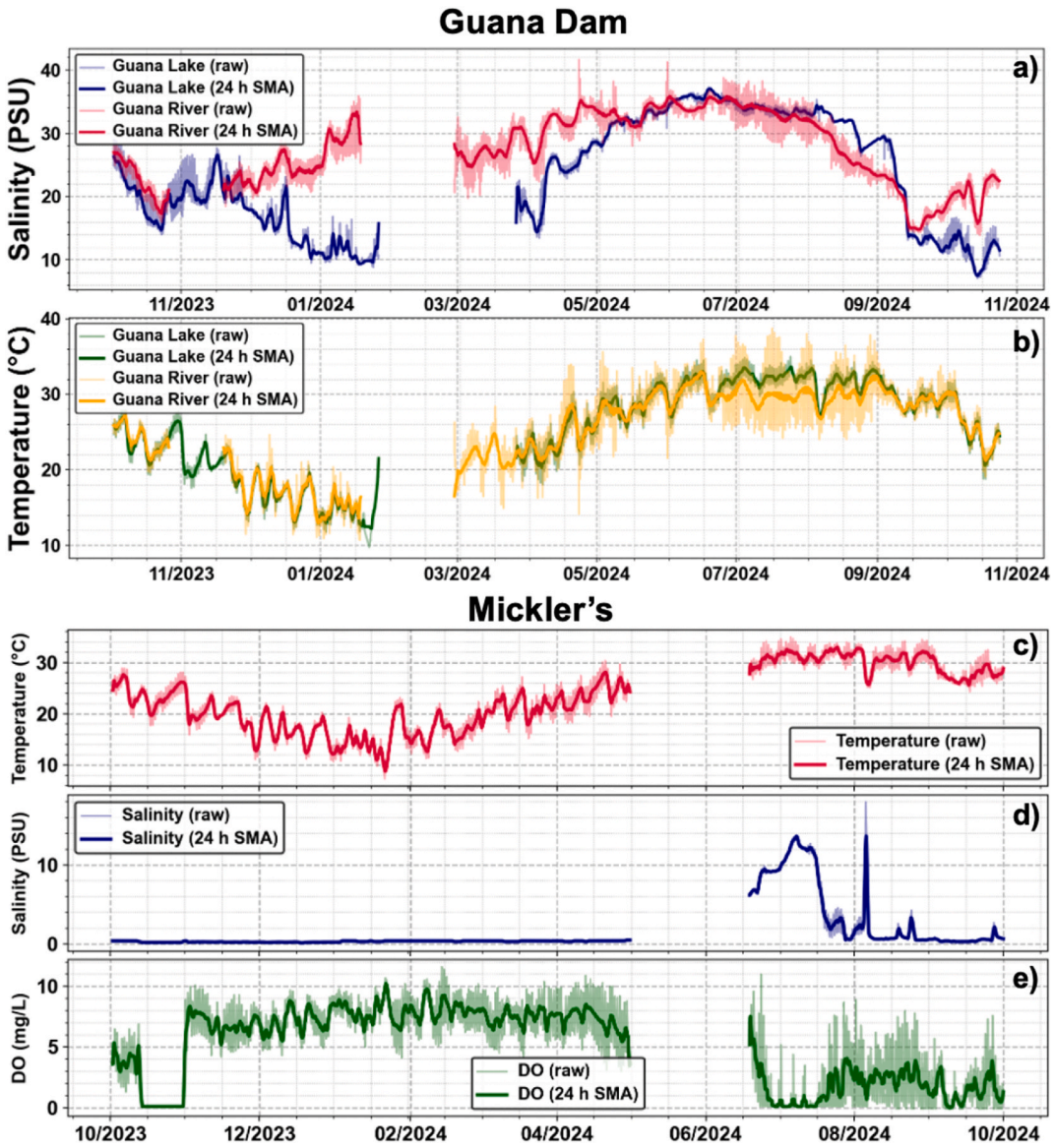


Fig. 4. (a): comparison between time series of salinity (HOBO) in Guana Lake (blue line) and Guana River (red line). (b): temperature time series (HOBO) for Guana Lake (green line) and the Guana River (yellow line). Following, time series of the YSI EXO2 placed at Mickler's Weir. (c) Temperature (red line, expressed in °C), while (d) and (e) show the salinity (blue line, expressed in PSU) and the Dissolved Oxygen (green line, expressed in mg/L), respectively. The thicker line, in all cases, represents the simple moving average (SMA) of 24-h windows.

generally associated with lower DO. Seasonally, pH and DO increased during cooler months when oxygen solubility was higher. In contrast, driven by photosynthetic CO_2 uptake throughout summer, pH increased during early bloom onset but subsequently declined due to microbial respiration and organic matter decomposition, resulting in DO depletion and acidification.

4. Discussions

4.1. Spatial and temporal variability in collected data

TN in Guana Lake was dominated by organic nitrogen and ammonium (NH_4^+), with nitrate and nitrite remaining negligible throughout the one-year monitoring period. Mid-summer peaks in particulate organic nitrogen (PN) and NH_4^+, reaching up to 3 mg L^{-1} and 0.64 mg L^{-1}, respectively, were most pronounced at the northern sites (MK and GL2), while concentrations at LM and LS remained consistently low (green and red lines in Fig. 6b and f). These increases occurred after a major salinity intrusion event associated with dam operation and reduced freshwater inflow. Elevated temperatures, vegetation die off, and reduced exchange may have contributed to nitrogen accumulation under saline conditions, but the relative importance of these processes was not directly measured in this study.

Similar responses have been documented in other brackish and managed lakes where organic nitrogen dominates and episodic salinity or drought events trigger transient nitrogen pulses (Grefe et al., 2024). Salinity intrusions have been associated in other systems with mobilization of sediment bound and dissolved organic nitrogen, as well as resuspension of organic rich sediments. (Lin et al., 2023). Likewise, in Bafa Lake (Türkiye), maintaining freshwater inflows and limiting saltwater intrusion decreased nitrogen concentrations by 5–25 %, illustrating how hydrologic management directly governs nitrogen retention and release (Kacikoc et al., 2025).

In Guana Lake, salinity increases likely mobilized sediment-associated nitrogen, elevating DON, PN, and NH_4^+ while keeping dissolved inorganic nitrogen (DIN) largely in its reduced form. The near

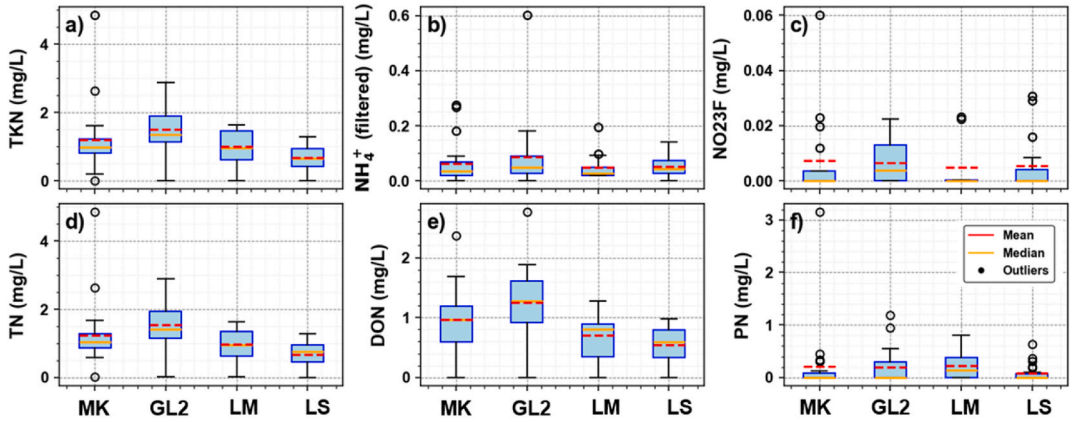


Fig. 5. Spatial variation of total nitrogen and its components during the period October 2023 and October 2024 in the four stations shown in Fig. 1b. (a) shows the Total Kjeldahl Nitrogen (unfiltered). (b) and (c) show the ammonium and NOx concentrations, both from filtered samples. (d) represents the total nitrogen, (e) Dissolved organic Nitrogen, and (f) Particulate Nitrogen.

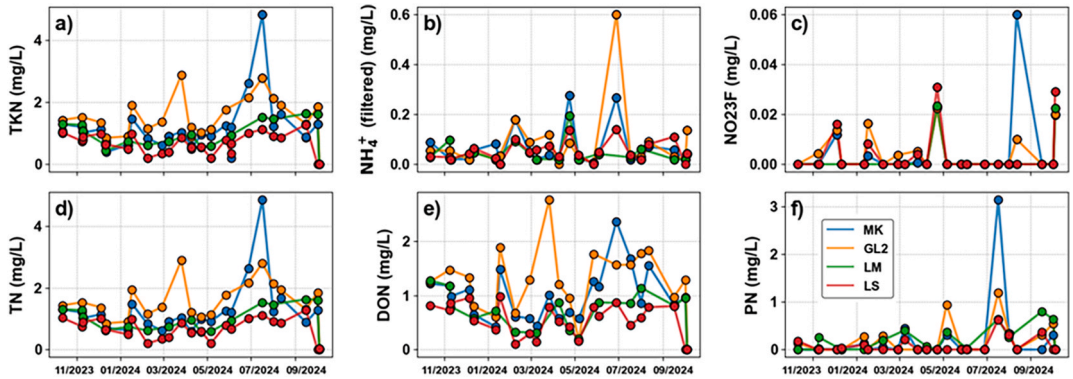


Fig. 6. Temporal variation in the total nitrogen and its component during the period October 2023, October 2024 in the four stations shown in Fig. 1b. (a) Total Kjeldahl Nitrogen (unfiltered). (b) and (c) show the ammonium and Nitrate + Nitrite concentrations, both from filtered samples. (d) represents the total nitrogen, (e) Dissolved Organic Nitrogen, and (f) Particulate Nitrogen.

Table 1
Correlation analysis for TN, salinity and temperature.

Station	n	r	p	Station	n	r	p	Station	n	r	p
TN-salinity				*TN-temperature*				*Temperature-salinity*			
GL2	20	0.369	0.11	**GL2**	20	0.084	0.726	**GL2**	20	0.692	0.001
LM	16	0.379	0.148	**LM**	16	0.619	0.011	**LM**	16	0.705	0.002
LS	23	0.009	0.969	**LS**	23	−0.01	0.957	**LS**	23	0.931	<0.001
MK	23	0.822	<0.001	**MK**	23	0.152	0.488	**MK**	23	0.434	0.039
NH_4^+ *-temperature*				*TN–DO*							
GL2	20	−0.34	0.143	**GL2**	20	−0.33	0.155				
LM	16	−0.651	0.006	**LM**	16	0.086	0.752				
LS	23	−0.243	0.263	**LS**	23	−0.18	0.401				
MK	23	−0.289	0.181	**MK**	23	−0.25	0.254				

absence of NOx indicates that nitrification was likely suppressed under warm, oxygen-poor, and saline conditions (Kacikoc et al., 2025). Another possible explanation is that oxidized nitrogen entering from upstream sources did not persist to the lake, either because of uptake or transformation before reaching the monitored site. As a result, nitrogen entering the system would be dominated by organic and reduced forms, consistent with the observed prevalence of DON, PN, and NH_4^+. Moreover, any oxidized nitrogen that did enter the system may have been rapidly assimilated by biota rather than persisting in the dissolved inorganic pool (Beman et al., 2013).

Peaks in TKN and TN at GL2 and LM coincide with salinity-driven nitrogen release, whereas upstream freshwater sites showed muted responses. These trends support the interpretation that salinity and temperature fluctuations are important associated factors of nitrogen cycling in this brackish reservoir, consistent with evidence that evapotranspiration-induced salinity spikes promote organic-N accumulation (Lin et al., 2023; Kacikoc et al., 2025).

Salinity also restructures the microbial and phytoplankton communities that mediate nitrogen transformations. Across brackish systems, higher salinity favors ammonia-oxidizing archaea over bacteria and shifts phytoplankton assemblages toward brackish-tolerant species with lower biomass (Zhang et al., 2015; Behera et al., 2017). These shifts have been reported to alter nitrogen processing in other brackish systems and may be consistent with the dominance of reduced nitrogen forms

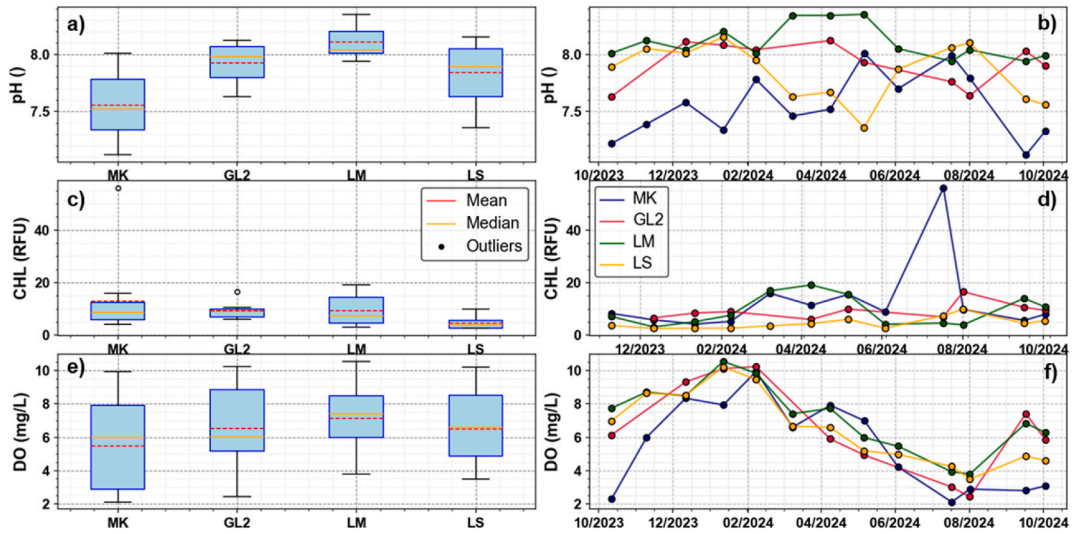


Fig. 7. Spatial and temporal variation in Guana Lake, respectively for: (a) and (b) pH, (c) and (d) Chlorophyll, (e) and (f) Dissolved Oxygen.

observed here.

Nitrogen fixation, though significant in some Florida lakes (Canham et al., 2012) and stormwater ponds (Goeckner et al., 2024), appears limited in the Lake. NOx remained below detection, and most TN occurred as organic nitrogen, suggesting that available nitrogen reduced the need for fixation. Some fixation may still occur, but any fixed nitrogen was likely assimilated immediately into biomass. Low dissolved oxygen during warm months further constrained nitrification–denitrification, hindering NH_4^+ oxidation and N_2 formation. Thus, the combination of high temperature and salinity is consistent with enhanced DON release and mineralization (Wang et al., 2021; Hocaoglu et al., 2011).

Biotic responses also reinforce these chemical patterns. Chlorophyll-a concentrations (Fig. 7d) increased only in the northern part of the Lake, coinciding with elevated temperature and salinity, indicating algal sensitivity to these conditions (Zhao et al., 2024). Similar responses have been observed in other brackish lakes, where salinity gradients influence phytoplankton composition, water transparency, and macrophyte growth (Stanković et al., 2024; Mrozińska et al., 2025). Rising temperatures accelerate algal metabolism and chlorophyll synthesis up to a threshold, beyond which thermal stress causes pigment degradation (Kholssi et al., 2023).

Overall, these results indicate that salinity, temperature, and dam-induced hydrologic variability coincided with nitrogen transformation and biological productivity in the Lake. The patterns observed align with findings from other saline and transitional systems, where physical forcing governs microbial community structure, nutrient recycling, and ecosystem functioning across spatial and temporal scales (Afonina et al., 2025). Taken together, the dataset shows that nitrogen accumulation was associated with periods of elevated salinity, high temperature, and altered gate operation, but it does not resolve whether the dominant cause was internal nitrogen release, external loading, reduced water exchange, or a combination of these processes.

4.2. Proposed conceptual model of nitrogen cycle in a managed estuary

The findings of this study highlight the central role of site-specific management, which affects nitrogen dynamics, with flow and salinity control emerging as key priorities. In particular, the strategy of introducing salt water to suppress nuisance freshwater vegetation was implemented by moving from a configuration with both discharge and recharge gates open to a configuration with no discharge gates and one recharge gate. This increased saltwater inflow and reduced the outflow, limited to a low discharge through the spillway, without maintaining bidirectional tidal flushing. Under these conditions, water exchange with the estuary was likely reduced, and nitrogen dynamics appear to have been influenced by both salinity changes and reduced outflow. The concurrent rise in lake level (Fig. 2c) is consistent with these recharge-only conditions, whereby saltwater entered while net outflow was minimized. On the one hand, increased salinity likely altered the nitrogen cycle and enhanced the release of particulate nitrogen through decomposition of freshwater vegetation. On the other hand, saltwater added from the estuary brought relatively low nitrogen concentrations and, in principle, diluted nitrogen in the Lake. Because TN increased after the intrusion event, the observations suggest that processes associated with saline intrusion and reduced exchange outweighed any simple dilution effect during this period.

Elevated salinity may contribute to die-off of nuisance plants, potentially liberating DON (Fig. 6e), and may enhance mineralization of organic nitrogen into NH_4^+. To prevent nuisance vegetation from reaching Guana Lake, mechanical harvesting could be implemented rather than relying on elevated salinity. However, mechanical harvesting tends to be more expensive (Antuniassi et al., 2002) than strategic dam and tidal-flow management.

High temperatures in summer further amplify microbial and enzymatic activity, accelerating nitrogen mineralization. This may elevate rates of NH_4^+ release even beyond the effects of salinity alone. In general, higher temperatures significantly boost NH_4^+ accumulation, with strong temperature sensitivity in wetland systems up to 40 °C (Gao et al., 2014). Meanwhile, increased salinity reduces ammonium adsorption and inhibits the activity of nitrifying and denitrifying bacteria, cutting NH_4^+ retention in the sediment (Rysgaard et al., 1999). The combination of high heat and salt thus likely drives DON release and mineralization while simultaneously hindering downstream nitrogen transformation pathways.

Alternatively, by operating gates to have both discharge and recharge within a tidal cycle, tidal flushing can be increased. This may help limit nitrogen accumulation by increasing exchange, although that hypothesis should be tested with hydrodynamic and water quality modeling. Scenario modeling for other controlled coastal lakes (e.g., Bafa Lake) showed that restoring more natural hydrologic conditions lowered nitrogen concentrations by shortening residence time and enhancing flushing (Kacikoc et al., 2025). However, in Bafa Lake there was not a need to drastically increase salinity levels and kill nuisance freshwater vegetation. If a similar strategy were applied to Guana Lake, the optimal combination of discharge and recharge operations at Guana Dam would need to be determined to balance nitrogen export, salinity targets for vegetation control, and downstream water quality (Fig. 8).

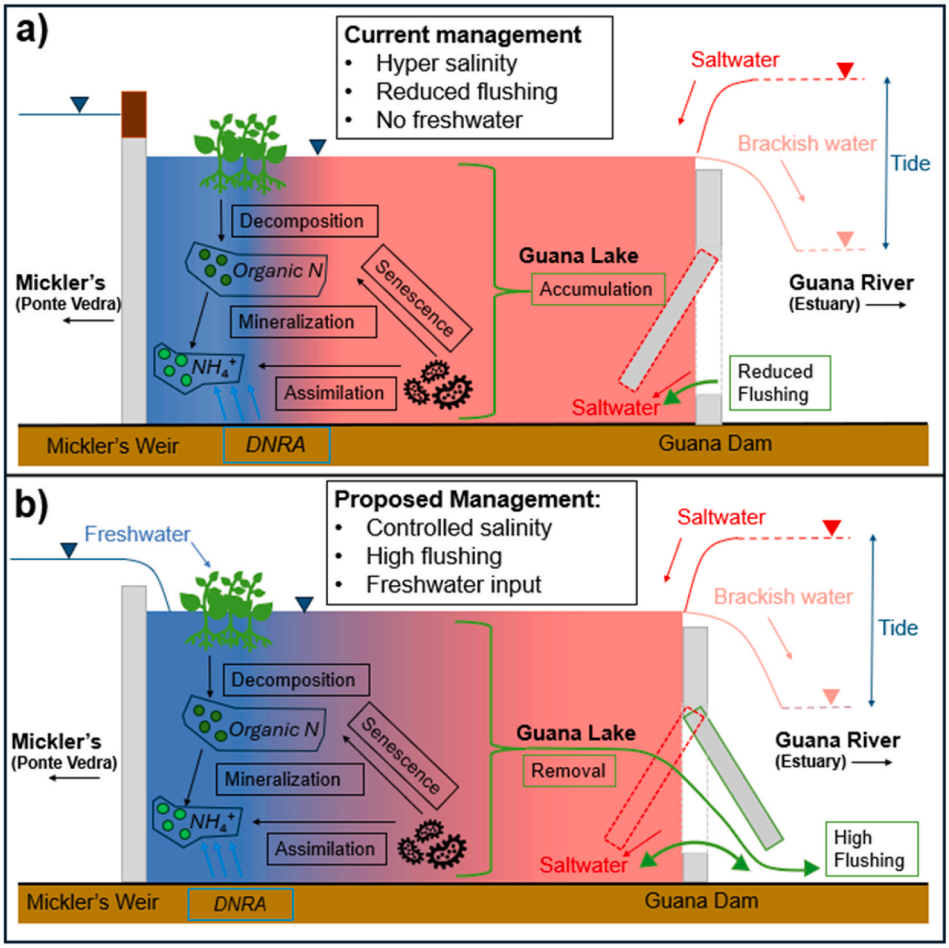


Fig. 8. Conceptual design of the proposed alternative management of Guana Lake. Panel (a) shows the current management: saltwater inflow has been increased to control freshwater vegetation, but discharge gates are closed, reducing outflow and tidal exchange, which leads to nitrogen accumulation under the mechanisms described in Section 4.1. Panel (b) illustrates the proposed management: a combination of recharge (dashed red) and discharge gates (green) re-establishes bidirectional tidal flushing, allowing excess nitrogen to be exported and preventing eutrophication.

This increase in flushing would also prevent damaging hypersaline conditions (seen in the moments where salinity was higher in the Lake than in the estuary, Figs. 3a and 4a). In addition to the Guana Dam management, ensuring year-round freshwater input at the Mickler's Weir would help stabilize salinity regimes and sustain nitrification–denitrification processes.

The influence of dam management on water quality highlights the value of hydrodynamic modeling to test alternative operational strategies. Such models could optimize the number of discharge and recharge gates needed, as well as their opening duration, to meet both flood control requirements and ecological thresholds in different seasons. More broadly, nitrogen reduction in brackish environments requires an integrated approach that combines hydrologic management with the restoration of vegetated zones, enhancement of nitrogen uptake and denitrification, and mitigation of external nitrogen loads.

Finally, numerical modeling and management decisions should account for the entire estuarine continuum. Although increased flushing in the Lake can improve in-lake water quality, it also transfers nitrogen downstream, potentially reducing the buffering capacity of the Lake for the estuary. Balancing lake-level water quality improvements with the risk of downstream nitrogen enrichment is therefore essential for system-wide management.

5. Conclusions

This study investigated the spatial and temporal variability of nitrogen concentrations in Guana Lake, a dam-managed brackish system in Northeast Florida. Motivated by concerns over declining water quality and episodic algal blooms, we conducted a one-year monitoring campaign to evaluate how dam operations and seasonal hydroclimatic variability relate to nitrogen dynamics. Monthly in situ measurements of temperature, salinity, dissolved oxygen (DO), pH, and chlorophyll-a were combined with laboratory analyses of total nitrogen (TN) and its constituent forms, including ammonium (NH_4^+), nitrate and nitrite ($NO_3^- + NO_2^-$), dissolved organic nitrogen (DON), and particulate nitrogen (PN).

The results indicate that periods of recharge-mode dam operation coincided with increased salinity, temperature, and water residence time, and were associated with elevated TN concentrations dominated by organic and reduced nitrogen forms. Elevated DON and NH_4^+ during these periods are consistent with enhanced organic matter decomposition and nitrogen mineralization under warm, saline, and low-oxygen conditions. Nitrate and nitrite remained below detection limits throughout the monitoring period, suggesting rapid uptake or transformation of oxidized nitrogen, although external nitrate inputs cannot be excluded. Reduced DO during summer months likely further constrained nitrification and denitrification pathways, allowing ammonium to persist in the water column.

While this study is observational and does not directly quantify hydrodynamic fluxes, residence time, or sediment nitrogen process rates, the consistent temporal and spatial correspondence between dam operations, salinity intrusion, and nitrogen accumulation provides

consistent observational evidence that hydrologic management plays a key role in shaping nitrogen dynamics in this system. Elevated TN concentrations at GL2 and Mickler's stations were closely associated with upstream nutrient inputs and episodic saltwater intrusion, consistent with responses observed in other transitional and regulated aquatic systems. The presence of a bidirectional dam at Guana Lake represents a distinctive control mechanism that offers opportunities for adaptive management of salinity and flushing regimes.

These findings suggest that hydrologic strategies that enhance tidal flushing may help mitigate nitrogen accumulation in brackish–freshwater systems. Increasing freshwater inflows during summer and modifying dam operations to promote exchange could reduce TN buildup and lower the risk of algal blooms. Such hydrologic management should be implemented alongside watershed nutrient source reduction, habitat restoration, and vegetation management to improve water quality and ecological resilience.

Future work should integrate hydrodynamic and biogeochemical modeling with targeted process measurements to quantify residence time, sediment nitrogen fluxes, and transformation pathways under alternative management scenarios. This will provide a mechanistic basis for optimizing dam operations to balance flood protection, vegetation control, and water quality objectives.

CRediT authorship contribution statement

Stefano Biondi: Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Daniele Pinton:** Writing – review & editing, Methodology, Conceptualization. **Chia Chu Chu:** Investigation. **Nikki Dix:** Writing – review & editing, Resources, Conceptualization. **Ashley Smyth:** Writing – review & editing, Supervision. **Alexander J. Reisinger:** Writing – review & editing, Supervision. **Alberto Canestrelli:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Stefano Biondi reports financial support was provided by University of Florida. Stefano Biondi reports a relationship with University of Florida that includes: employment. Stefano Biondi has patent pending to n/a. Given his role as, had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2026.129590.

Data availability

Data will be made available on request.

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